Assignment11 – Natural Language

Given: July 1   Due: July 6

Problem 11.1 (Language Models)

Consider a language L over the alphabet with characters a, b, c, d, and consider a corpus D for it consisting of the five words

$$a\,b\,c\,c\,c \qquad a\,b\,b\,d\,a \qquad a\,a\,a\,b\,c \qquad b\,c\,d\,b \qquad a\,a\,b\,a\,c$$

You want to build a 3-gram model for L.

1. How many 3-grams are there over L?

   Solution: $4^3 = 64$

2. Now assume you store the 3-gram model as a Bayesian network with variables w_1, w_2, and w_3 (representing the 3-gram $w_1\,w_2\,w_3$), how many entries does the conditional probability table for w_3 have?

   Solution: 444 = 64 or 443 = 48 if we utilize redundancy. Either answer was accepted.

3. Give the numerical value of $P(w_3 = a \mid w_{1:2} = aa)$ in the resulting model.

   Solution: $1/3$

4. Give the numerical value of $\mathrm{tf}(a, cdad)$.

   Solution: $\frac{1}{4}$

5. Give the numerical value of $\mathrm{idf}(a, D)$.

   Solution: $\log_{10} 5/4$

6. Now you want to use the tfidf vector embedding. What are the objects that are embedded, and how are the vectors computed using tf and idf?

Solution: We embed documents/words W from L. The vector for W is $\mathrm{tf}(x, W) \cdot \mathrm{idf}(x, D)$ for $x \in \{a, b, c, d\}$.

7. Now assume you are interested in all documents that contain the subword $b\,c$. A query system returns all 5 documents.
   Give the numerical values of
   1. precision: $\boxed{3/5}$
   2. recall: $\boxed{1}$

Solution: filled in above

Problem 11.2 (Information Retrieval)

1. Explain (in about 2 sentences) the connection between tfidf and cosine similarity to relate texts.

Solution: tfidf is a way to represent a text as a vector, relative to a corpus, by measuring the frequency of terms in some way. Cosine similarity uses the angle between two such vectors as a measure of how similar the texts represented by the vectors are.

2. Explain (in about 2 sentences) the difference between information retrieval and information extraction.

Solution: Information retrieval identifies documents that are likely to contain information that answers the query. Information extraction obtains structured representations of parts of the content of a document, which can then be used to answer queries precisely.

3. Explain (in about 3 sentences) what kind of function is learned in the continuous bag of words algorithm, as used e.g. in Word2Vec, and how the training proceeds.

Solution: It learns a word embedding, i.e., a function from words to vectors. Going through a corpus, each word is processed together with the context, i.e., the n words before and after. The neural network is trained to predict the word from its context.

Problem 11.3 (Grammars)

Consider the following probabilistic grammar:

$$
\begin{array}{lll}
S & \rightarrow & NP\ VP[1] \\
NP & \rightarrow & Article\ Noun[0.6] \mid Name[0.4] \\
VP & \rightarrow & Verb[0.5] \mid TransVerb\ NP[0.5] \\
Article & \rightarrow & \text{the}[0.7] \mid \text{a}[0.3] \\
Noun & \rightarrow & \text{stench}[0.2] \mid \text{breeze}[0.3] \mid \text{wumpus}[0.5] \\
Name & \rightarrow & \text{John}[0.3] \mid \text{Mary}[0.7] \\
Verb & \rightarrow & \text{smells}[1] \\
TransVerb & \rightarrow & \text{sees}[0.6] \mid \text{shoots}[0.4]
\end{array}
$$

1. Using this grammar as an example, explain the difference between structurals rule and lexicon.

Solution: Both are productions of the grammar. Structural rule define the language in general (S to $Article$, above), the lexicon defines the specific words used in a context ($Noun$ to $TransVerb$ above).
Note that the rules for $Article$ are a gray area: The distinction between the and a can be seen as a matter of grammar or lexicon. If the former is intended, it would be clearer to introduce intermediate non-terminals like $DetArticle$ and $IndetArticle$. Then $Article$ would have structural rules, and $DetArticle$ and $IndetArticle$ lexical ones.

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2. Give the probability of the sentence

   John sees the wumpus

   (You have to give the expression with concrete values plugged in, but you do not have to compute the result.)

Solution: $0.4 \cdot 0.3 \cdot 0.5 \cdot 0.6 \cdot 0.6 \cdot 0.7 \cdot 0.5 = 0.00756$

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3. Now assume we do not know the probabilities of the production, and our corpus is

   John sees the wumpus. The wumpus smells. John shoots the wumpus.

   Give the probability that we can learn for $NP \rightarrow Name$ from this corpus.

Solution: We count all subtrees of type NP (N) and among those the ones of type $Name\ (n)$, then we learn the probability n div N. That yields $\frac{2}{5}$.
